

## Pre-Board Examination - 2079

**Class : XII**

**Subject: Chemistry (3021)**

**Full Marks: 75**

**Time: 3 hrs.**

**SET 'A'**

**Pass Marks: 30**

*Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.*

**Attempt all the questions.**

### Group 'A'

**Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]**

1. The solubility product of  $\text{CaF}_2$  is  $3.4 \times 10^{-11}$ . What is its solubility in 0.01M solution of NaF?
 

a)  $3.4 \times 10^{-7} \text{ mol l}^{-1}$

c)  $3.4 \times 10^{-2} \text{ mol l}^{-1}$

b)  $3.4 \times 10^{-5} \text{ mol l}^{-1}$

d)  $3.4 \text{ mol l}^{-1}$
2. Which of the following conditions are correct for spontaneous change?
 

a)  $\Delta H = 0, \Delta S < 0$

c)  $\Delta H < 0, \Delta S < 0$

b)  $\Delta H < 0, \Delta S > 0$

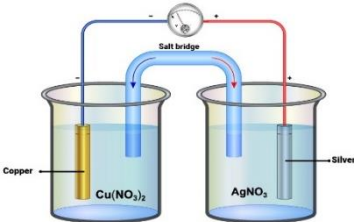
d)  $\Delta H > 0, \Delta S < 0$
3. For a galvanic cell given in the figure. Which of these isn't true?
 

a) Ag-electrode is cathode

b) Cu-electrode is anode

c)  $\text{Cu}^{++}$  is reduced

d)  $\text{Ag}^+$  is reduced


4. For rate law experiment, two gases (X) and any (Y) are filled in a container. The rate law for the reaction has been found to be,  $\text{rate} = k[\text{X}]^2[\text{Y}]$ . What will be the rate of reaction when pressure of both (X) and (Y) are doubled?
 

a) The rate will be doubled

c) The rate will becomes six times

b) The rate will becomes four times

d) The rate will becomes eight times

- Which of the following characteristics of transition metals is associated with their catalytic activity?  
a) Variable oxidation states  
b) Magnetic behavior  
c) Colour of ion  
d) d-d transition
- Bell metal is an alloy of  
a) Cu, Pb and Sn  
b) Sn and Cu  
c) Zn and Pb  
d) Zn, Cu and Sn
- (A)  $\xrightarrow{aq.NaOH}$  (B)  $\xrightarrow{K_2Cr_2O_7/H^+}$  (C).  
The compound (C) undergoes Clemmenson's reduction to give propane. The compound (A) is  
a) Propan-1-ol  
b) Propan-2-ol  
c) 2-halo propane  
d) Propanoic acid
- Bromination of aniline gives  
a) m-bromoaniline  
b) o-bromoaniline  
c) o,p- dibromoaniline  
d) 2,4,6 - tribromoaniline
- Which of the following reacts with alc. KCN to form benzoin.  
a)  $C_6H_5COOH$   
b)  $C_6H_5Cl$   
c)  $C_6H_5CHO$   
d)  $C_6H_5OH$
- An organic compound (X) undergoes reduction with  $LiAlH_4$  to yield (Y). When vapour of (Y) are passed over heated copper at  $300^\circ C$ , (X) is formed. The compound (Y) is  
a)  $CH_3CHO$   
b)  $CH_3CH_2OH$   
c)  $CH_3COCH_3$   
d)  $CH_3OCH_3$
- Which of the following is not an organometallic compound?  
a)  $CH_3ONa$   
b)  $C_2H_5Li$   
c)  $CH_3MgBr$   
d)  $(C_2H_5)_4Pb$

### Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. Concentration of sulphuric acid solution can be determined by titration with primary standard sodium carbonate solution.
  - a.  $\text{Na}_2\text{CO}_3$  is considered as primary standard why? [1]
  - b. Give reason why aqueous solution of  $\text{Na}_2\text{CO}_3$  is alkaline. [1]
  - c. 2.16 g of pure  $\text{Na}_2\text{CO}_3$  is added to 400 ml deci-normal solution of  $\text{H}_2\text{SO}_4$ . How many grams of  $\text{H}_2\text{SO}_4$  is further required to neutralize the resultant solution completely? [3]
2. How is order of reaction different from molecularity of reaction? [2]

The following rate data were obtained for the reaction



| Exp. No. | [A] mol L <sup>-1</sup> | [B] mol L <sup>-1</sup> | Initial rate of formation of C mol L <sup>-1</sup> |
|----------|-------------------------|-------------------------|----------------------------------------------------|
| 1        | 0.1                     | 0.1                     | $6.0 \times 10^{-3}$                               |
| 2        | 0.3                     | 0.2                     | $7.2 \times 10^{-2}$                               |
| 3        | 0.3                     | 0.4                     | $2.88 \times 10^{-1}$                              |
| 4        | 0.4                     | 0.1                     | $2.4 \times 10^{-2}$                               |

Determine:

- a. Overall order of reaction [2]
- b. Rate of formation of C when  $[\text{A}] = 0.5 \text{ mol L}^{-1}$  and  $[\text{B}] = 0.2 \text{ mol L}^{-1}$  [1]

**OR**

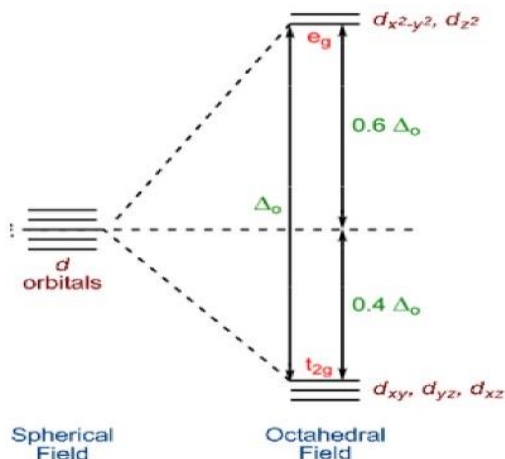
Hess's law is applied to calculate the different types of enthalpy of reactions.

- i. Define Hess's law of constant heat summation. [1]
- ii. What is meant by enthalpy of reaction? [1]

- iii. Calculate heat of formation of ethyl alcohol from the given data. [3]

|                                     |            |
|-------------------------------------|------------|
| Heat of combustion of ethyl alcohol | -330 KCal  |
| Heat of formation of carbondioxide  | -94 KCal   |
| Heat of formation of water          | -68.5 KCal |

3. a. Why do transition elements show variable oxidation states? Explain with reference to 3d series. [2]
- b. The figure shows the octahedral distortion of d-orbital in the presence of ligand.



- i. Why does octahedral distortion occur in the presence of ligand? Explain on the basis of CFT. [2]
- ii. On the basis of the given distortion, how can you explain  $[\text{Cu}(\text{H}_2\text{O})_6]^{++}$  is blue colored complex. [1]
4. a. Steel is the most important form of commercial iron. Discuss the basic oxygen process (BOP) for manufacture of steel. [3]

- b. Zinc is considered as non-typical element and it belongs to the element of group II B.
- i. Name the process of concentration of Zinc blende during the extraction of Zn. [1]
- ii. Write chemical reaction involved in roasting during extraction of zinc. [1]
5. You are given an organic compound having molecular formula  $C_3H_7Cl$ .
- a. Write down a primary and a secondary haloalkane giving their IUPAC name. [1]
- b. Give proper reaction sequence to convert the above primary haloalkane into the secondary haloalkane. [2]
- c. What is  $SN^1$  reaction? Which of the above isomers have higher reactivity towards  $SN^1$  mechanism? [1]
- d. Give reason, why haloalkanes are more reactive than haloarenes towards nucleophilic substitution reaction. [1]
6. Nitrobenzene is an industrial chemical and yellow oily liquid with almond like odour.
- a. Write down the chemical reaction for the nitration of benzene to prepare nitrobenzene. [1]
- b. How would you convert nitrobenzene into (i) para-aminophenol (ii) N-phenyl hydroxyl amine (iii) azoxybenzene [3]
- c. Give a structural formula of an aromatic poly nitro compound which is used as explosives. [1]
7. Ethers have wide use as commercial solvents and extractants for organic substances like esters, gums, hydrocarbons, alkaloids, oils, resins, dyes, plastics, lacquers, and paints.
- i. How would you prepare diethylether from  $C_2H_5Br$  by using Williamson's reaction?

- ii. What product is obtained when ethoxyethane is heated with excess HI?
- iii. Why are old sample of ether not distilled to dryness?
- iv. Convert phenol into anisole. [1+1+2+1]

**OR**

Mention any two suitable method of preparation of primary amine. How would you separate primary, secondary and tertiary amines by Hoffmann's method? [2+3]

8. Write an example of each of the following reaction. [1×5 = 5]
- i. Reimer Tiemann's reaction
  - ii. Oxo process
  - iii. Aldol condensation
  - iv. Tollen's test
  - v. HVZ reaction

**Group - C**

**Give long answer to the following questions.**

**[3 × 8 = 24]**

9. a. Standard hydrogen electrode is an electrode that scientists use for reference on all half-cell potential reactions.
- i. What is meant by reference electrode? [1]
  - ii. For the measurement of standard electrode potential of silver  $E^{\circ}_{\text{Ag}^+/\text{Ag}}$ , Hydrogen electrode is connected with it.
    - I) Represent an electrochemical cell notation.
    - II) Indicate cathode and anode.
    - III) If standard EMF of the cell ( $E_{\text{cell}}$ ) is 0.80 V, calculate the standard electrode potential  $E^{\circ}_{\text{Ag}^+/\text{Ag}}$ . [3]

- b. Discuss the criteria of spontaneity, non-spontaneity and equilibrium of exothermic and endothermic reactions on the basis of free energy and entropy change. [4]

**OR**

- a. What amount of  $\text{Zn(OH)}_2$  will be precipitated out at  $25^\circ\text{C}$  if 100 ml of 0.22gm NaOH is added to 1 liter of a saturated solution of  $\text{Zn(OH)}_2$ ? [Solubility product of  $\text{Zn(OH)}_2$  at  $25^\circ\text{C}$  is  $1.8 \times 10^{-14}$ .] [4]
- b. For a hypothetical reaction ( $\text{P} + 2\text{Q} \rightarrow \text{M} + \text{N}$ ), Concentration of (M) is increased by  $0.02 \text{ ML}^{-1}$  in 10 seconds. [4]
- Define rate of reaction.
  - Write down rate expression for the above reaction.
  - Calculate the rate of formation of (M).
  - If the rate constant of first order reaction is  $2 \times 10^{-3} \text{ s}^{-1}$ , what will be the half-life of this reaction?

10. a. The given table shows the different organic compounds.

|                                                                |                                                                        |
|----------------------------------------------------------------|------------------------------------------------------------------------|
| $\text{A} \Rightarrow \text{C}_6\text{H}_5\text{Cl}$           | $\text{D} \Rightarrow \text{C}_6\text{H}_5 \text{CH} = \text{CH COOH}$ |
| $\text{B} \Rightarrow \text{C}_6\text{H}_5\text{CHO}$          | $\text{E} \Rightarrow \text{C}_6\text{H}_5 \text{CH}_3$                |
| $\text{C} \Rightarrow \text{C}_6\text{H}_5\text{N}_2\text{Cl}$ | $\text{F} \Rightarrow \text{C}_6\text{H}_5 \text{NH}_2$                |

Make a correct reaction sequence of the above compounds with proper conditions and reagent required. [6]

- b. Why does nitrobenzene give meta substituted product during electrophilic substitution reaction? [2]

**OR**

- a. Write down correct reaction of each for the preparation of acetaldehyde from following sets of organic compounds.
- i.  $\text{CH}_3 \text{COCl}$
  - ii.  $\text{CH}_3 \text{CH}_2 \text{OH}$
  - iii.  $\text{CH}_3 \text{CHCl}_2$  [3]
- b. Starting from acetaldehyde how would you obtain lactic acid? [2]
- c. Write the primary and secondary amines from molecular formula  $\text{C}_2\text{H}_7\text{N}$ . Give a suitable chemical test to distinguish these isomers. [1+2]

11. a. Cement is important materials for the constructions of building, road, bridge etc.
- i. Give a difference between hydraulic and non-hydraulic cement. [2]
  - ii. List the compositions of Portland cement. [1]
- b. What is meant by artificial radioactivity? Write an example of it. [2]
- c. PVC and nylon-6,6 are two common polymers widely used in daily life. [2]
- i. State the process of polymerization by which nylon-6, 6 is formed.
  - ii. Write down the chemical reaction to form PVC.
- d. Define pulp and paper. [1]

**❧ - The End - ❧**